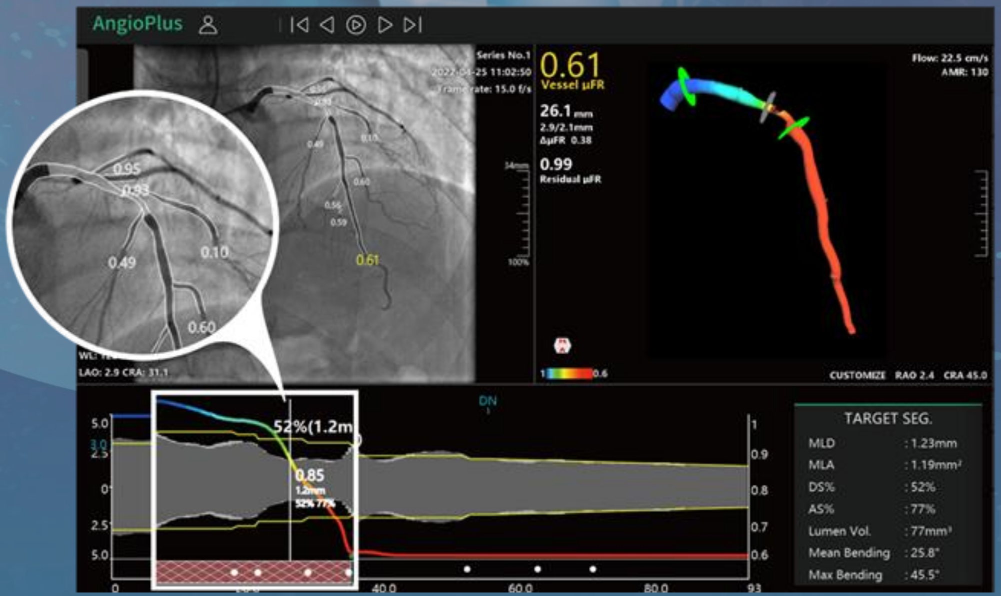

μFR® (AngioPlus Core)

--Angio-based FFR and PCI Planning

Pulse Medical

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µFR analysis process

Angiography | **Image selection**: Recommended projection, clear angiography, less overlapping, less foreshortening

Flow	2D µFR	2^{nd} View	3D µFR
• Check flow velocity • Select key frame • Select ROI (Region Of Interest) • Check vessel contour	• Branch detection • Check reference lumen	• Select key frame • Select ROI • Check vessel contour • Corresponding marks	• Check 3D reconstruction morphology • Check pressure pull back curve • Check diameter change curve • Check reference lumen

Export 2D analysis report

Export 3D analysis report

SOP of Coronary Angiographic Image Acquisition

1. **Acquiring the angiographic image using recommended projection**

   Acquire **at least 1 angiographic image** runs according to the **recommended angiographic projections**. Adjust angiographic projections if the recommended projection has a suboptimal quality or vessel overlaps at the lesion segment.

2. **Uniformly injection, fill the entire vessel**

   keep catheter to have better co-axiality with respect to vessel and maintain the uniformly contrast injection speed (~4 ml/s) until the distal vessel is filled with contrast and keep it to include the ED phase.

3. **Nitroglycerin required**

   Inject nitroglycerin through the imaging/guiding catheter to **eliminate arterial spasm** prior to angiography.

4. **Angiographic frame rate ≥15 frame/s, start recording before contrast injection, ≥1 cardiac cycle**

   Set angiographic imaging frame rate **≥15 frame/sec**. Then prefill the catheter with contrast medium. After complete wash out of the contrast medium, Start recording angiographic images at least **1 second prior to contrast injection**. The angiogram should include **≥1 cardiac cycle**, preferably 2 cycles.

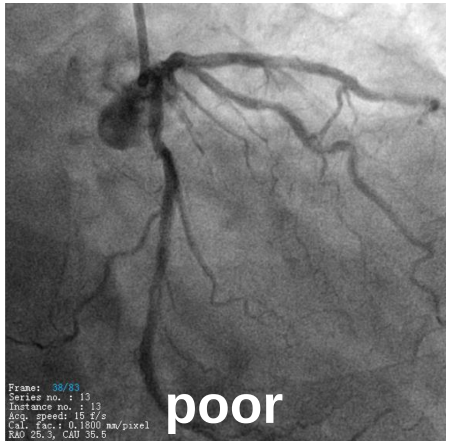
5. **Table moving**

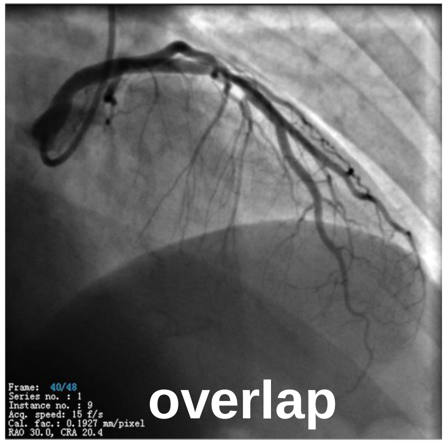
   Don’t move the table, especially the height.

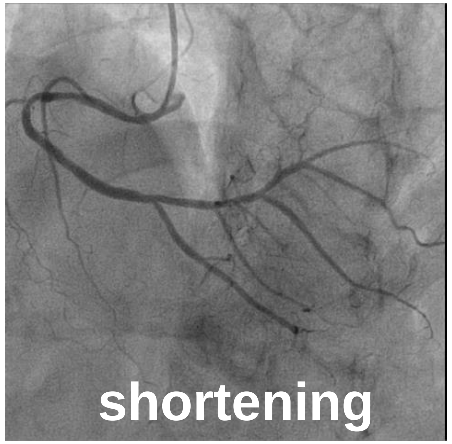
2D Analysis - Six key principles for qualified μFR analysis

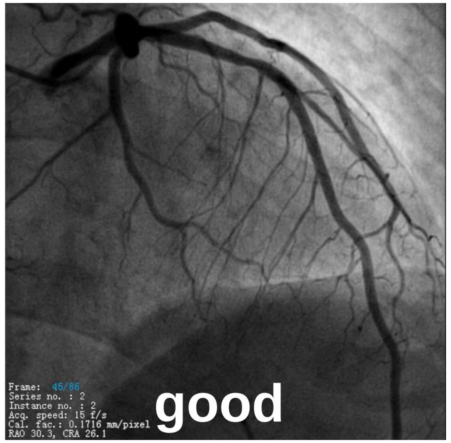
Image Selection	• Recommended projection, clear angiography, less overlapping, less foreshortening
Key Frame	• Prioritize the frame with the maximal stenosis exposure • Select the frame with the clearest vessel segment
Vessel Segmentation	• sufficient length (≥20mm), start point and end point in normal segment, moderate correction of contours
Flow Velocity	• Make sure the correct tracing of contrast/blood flow
Bifurcation	• Double check the detection and reference index position of bifurcations
Reference Lumen	• Ensure PN and DN in the positions with healthy lumen

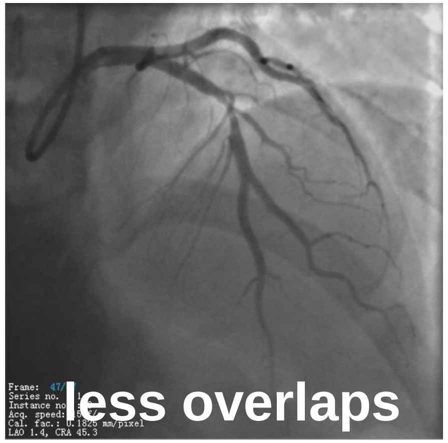
2D Analysis - Image selection

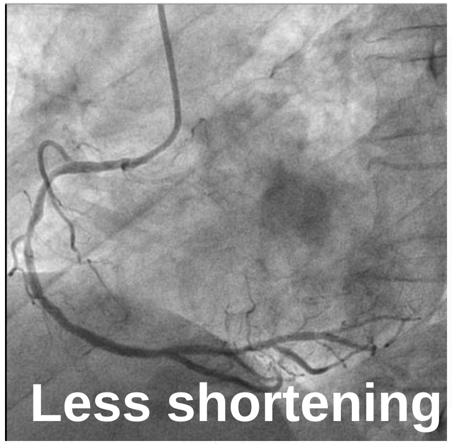










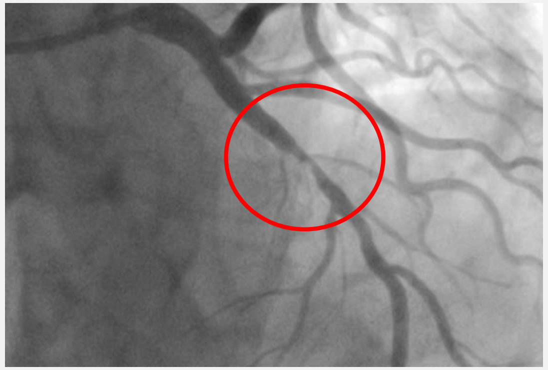


Note: Preferably choose optimal angiography runs (especially in lesion segment) with less vascular overlaps and less vascular shortening.

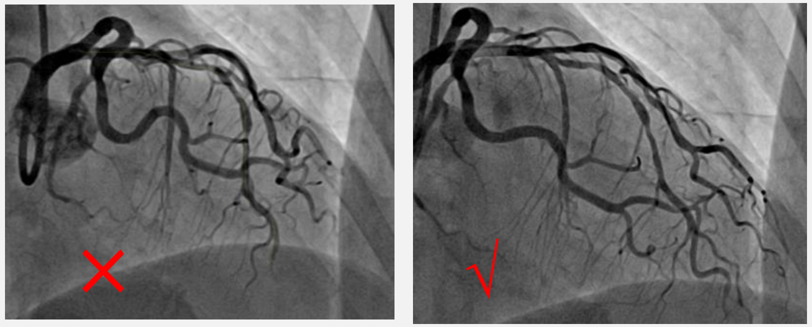
2D Analysis - Key Frame

Degree of importance : A ＞ B = C ＞ D

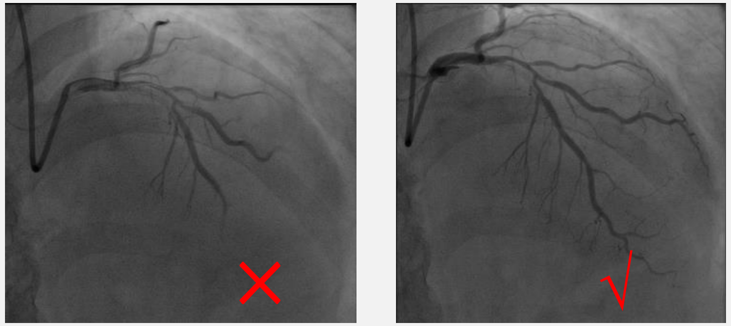
A Maximal stenosis exposure



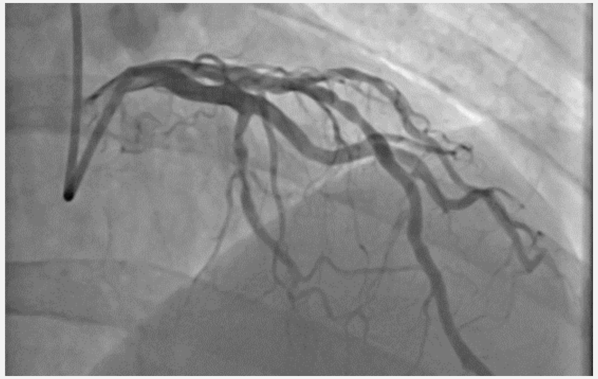
B Lower foreshortening



C Entire vessel clearly visualized



D Less overlapping



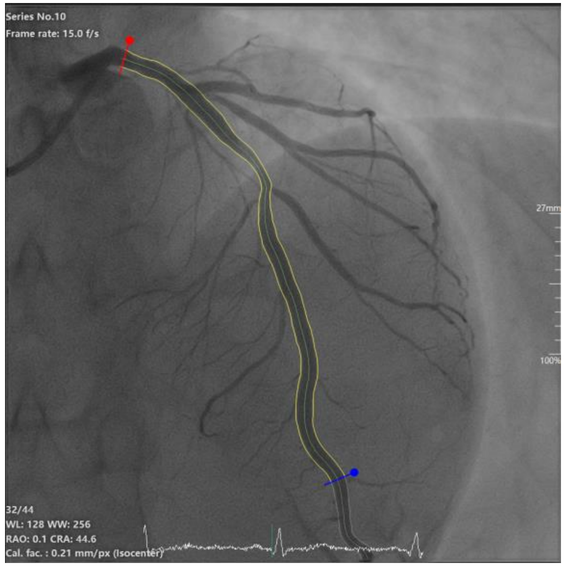
2D Analysis - Vessel Segmentation

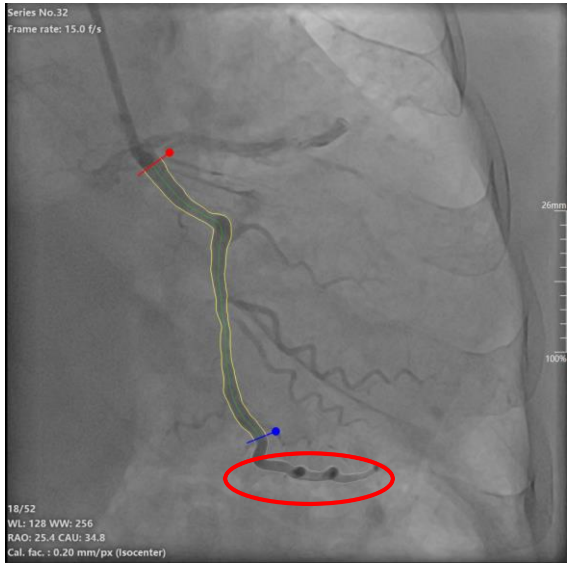
Length: ≥20mm and includes all lesions

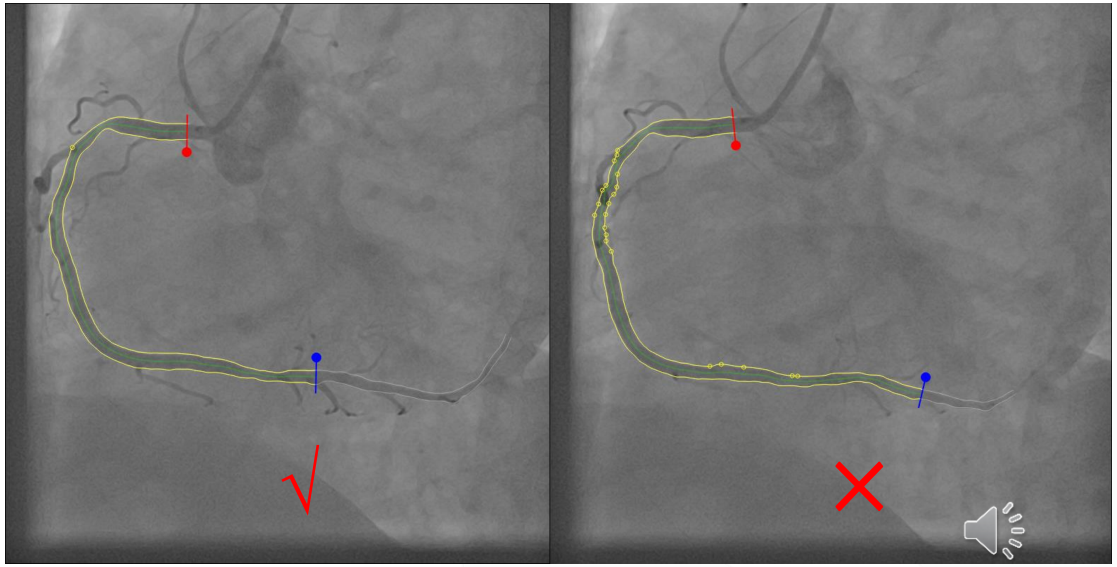
Start point: LM (LCA), beginning of the catheter (RCA) --(unless it is unclear and foreshortening or overlapping)

End point: normal segment with a marked point, exclude distortions

Vascular contours: good contrast- little correction; bad contrast- moderate correction



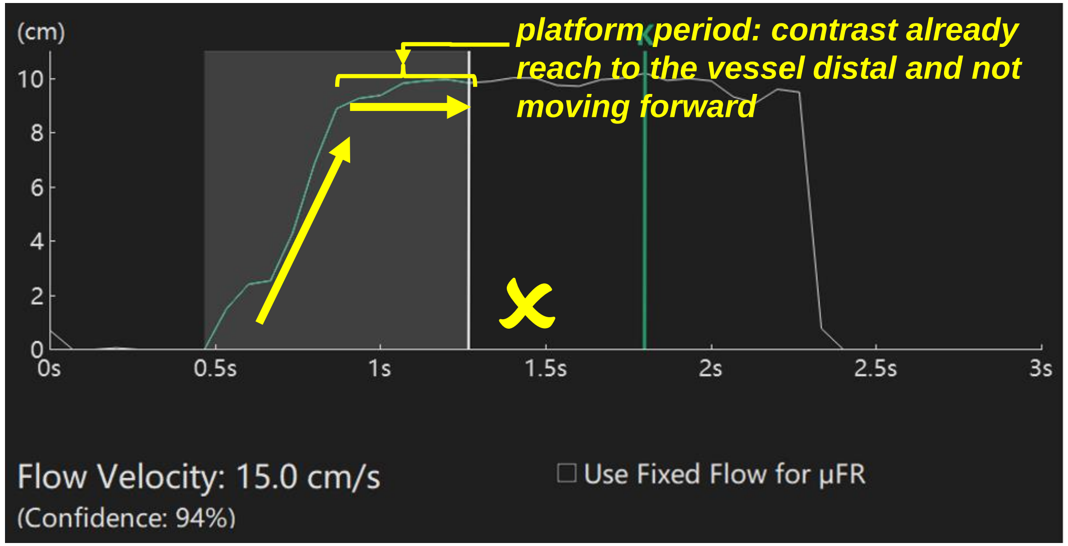


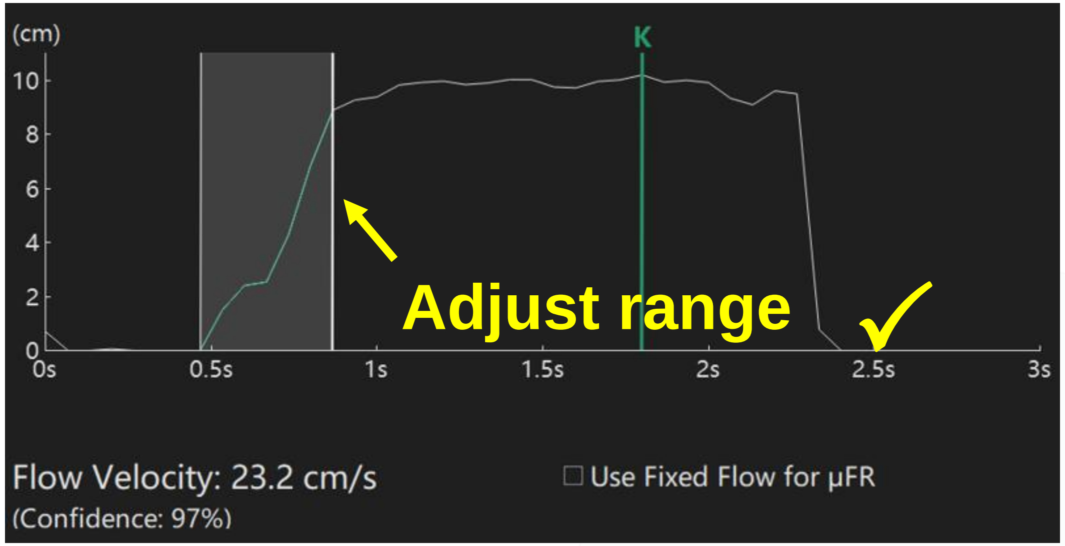


2D Analysis - Flow velocity

Check if the flow velocity curve, and adjust the range if necessary

- The grey box should cover all the frames, from the contrast entering the vessel, to the contrast flowing to the distal of vessel





Contains the period that contrast no longer flow further in the vessel (*platform period,* centerline length no longer increases)

correct flow velocity

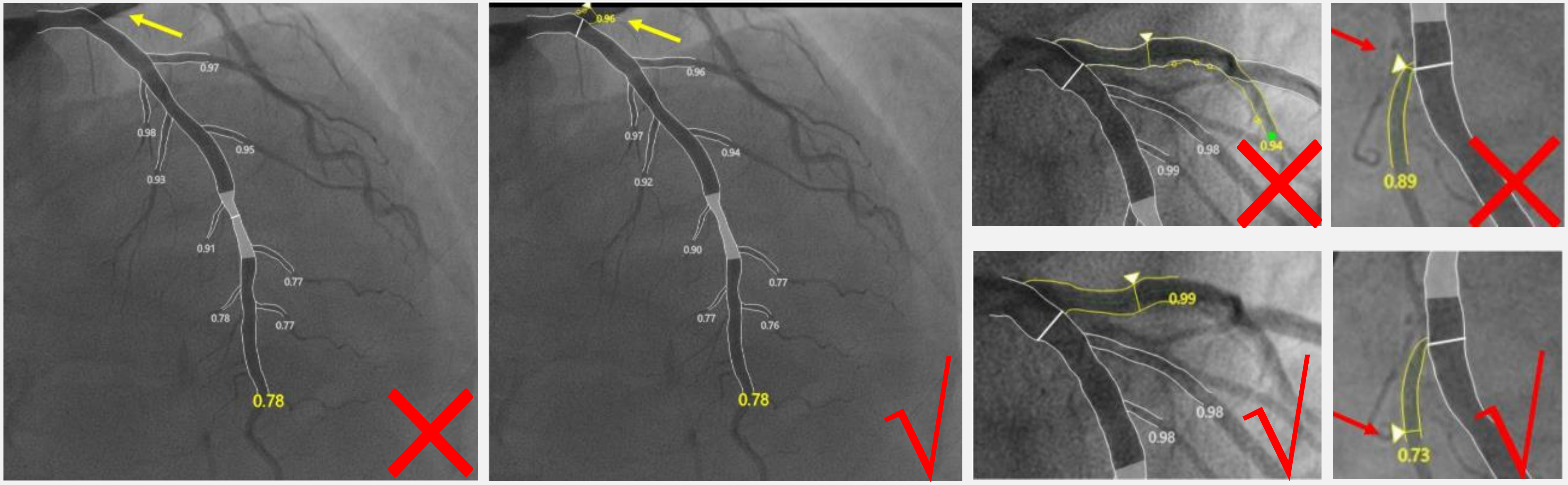
2D Analysis - Flow velocity

If the centerline cannot be accurately detected but you still want to analyze, then the fixed flow model can be applied. But the μFR results obtained from the fixed flow model should be regarded as for reference only.

2D Analysis - Bifurcation

Detection: identified all bifurcations (diameter≥1.0mm): manually add missing branches, delete excess branches. Put the endpoint of branch after the last stenosis.

Reference Diameter: reference index should be in healthy segment.

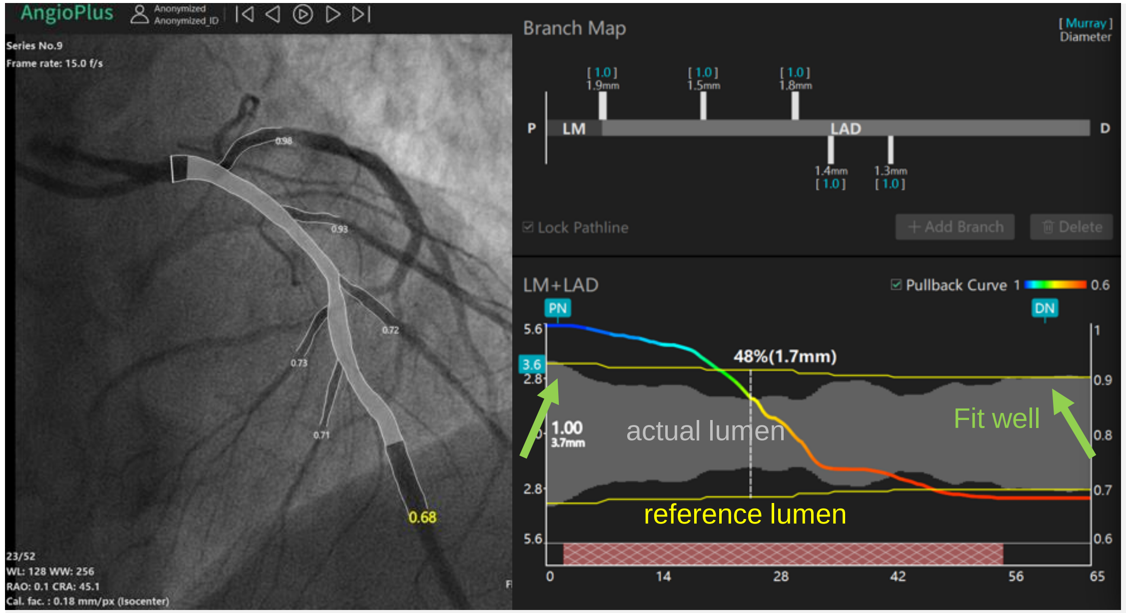


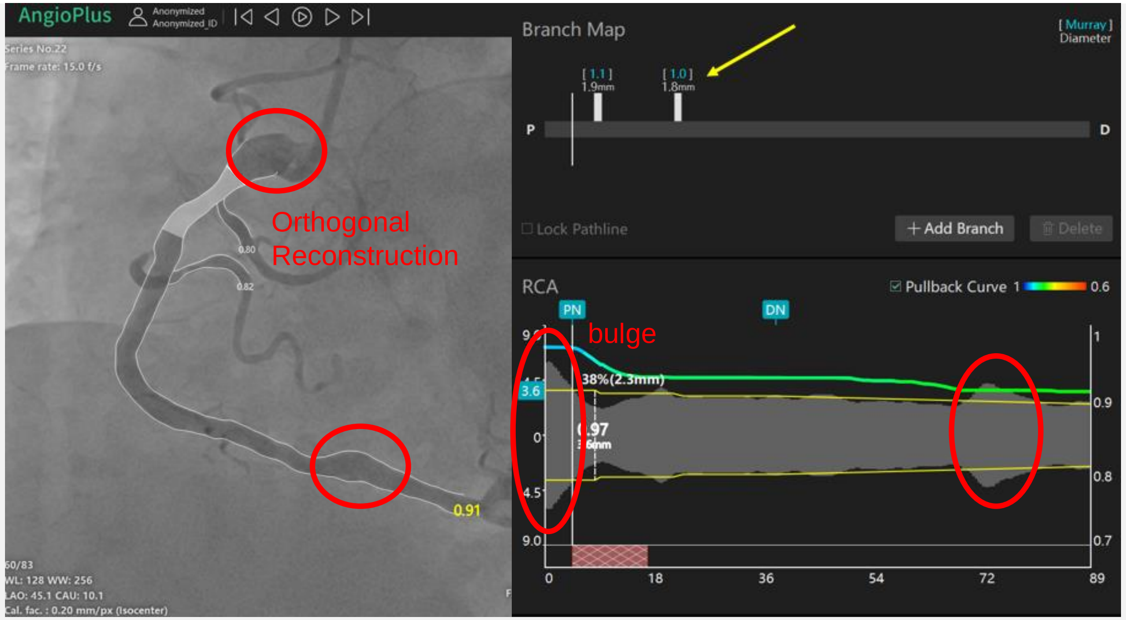
Missed the big side-branch **Wrong endpoint** **Wrong reference index**

2D Analysis - Reference Lumen

PN -proximal normal   DN -distal normal

Priority use of the automatic method





Fit well: reference lumen of the normal vessel segment fits the actual lumen **(Priority guarantee the normal segment adjacent to culprit lesion)**

PS: orthogonal reconstruction vessel will expand than normal segment

2D Analysis - Reference Lumen

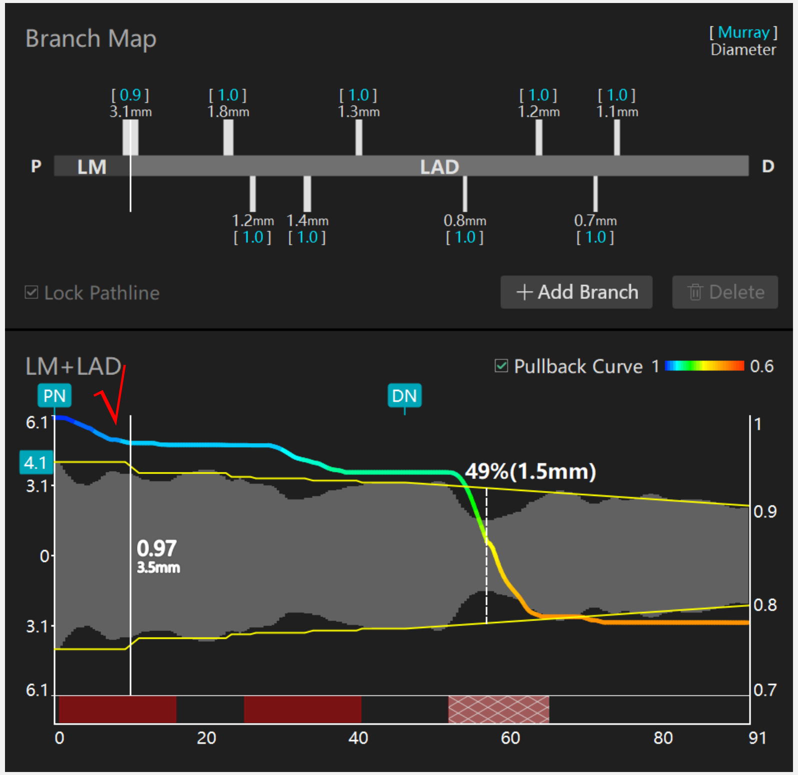
PN -proximal normal DN -distal normal

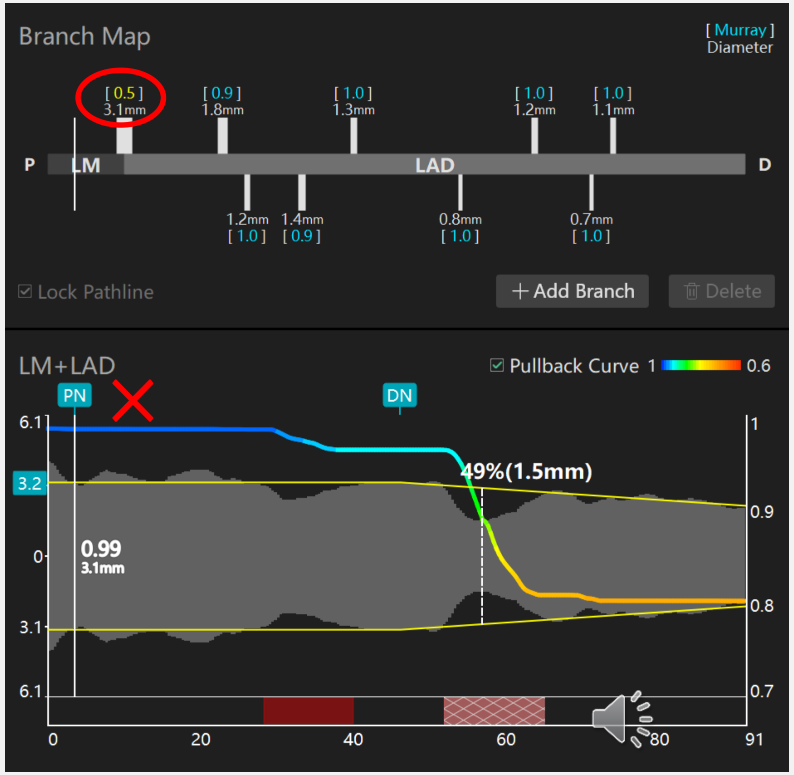
Priority use of the automatic method

Mu (μ) values

1. Make sure all the large side branches are recognized correctly.

2. Check Mu (μ) values are within the normal range **0.7 - 1.3**.





3D Analysis - Four key principles for qualified μFR analysis

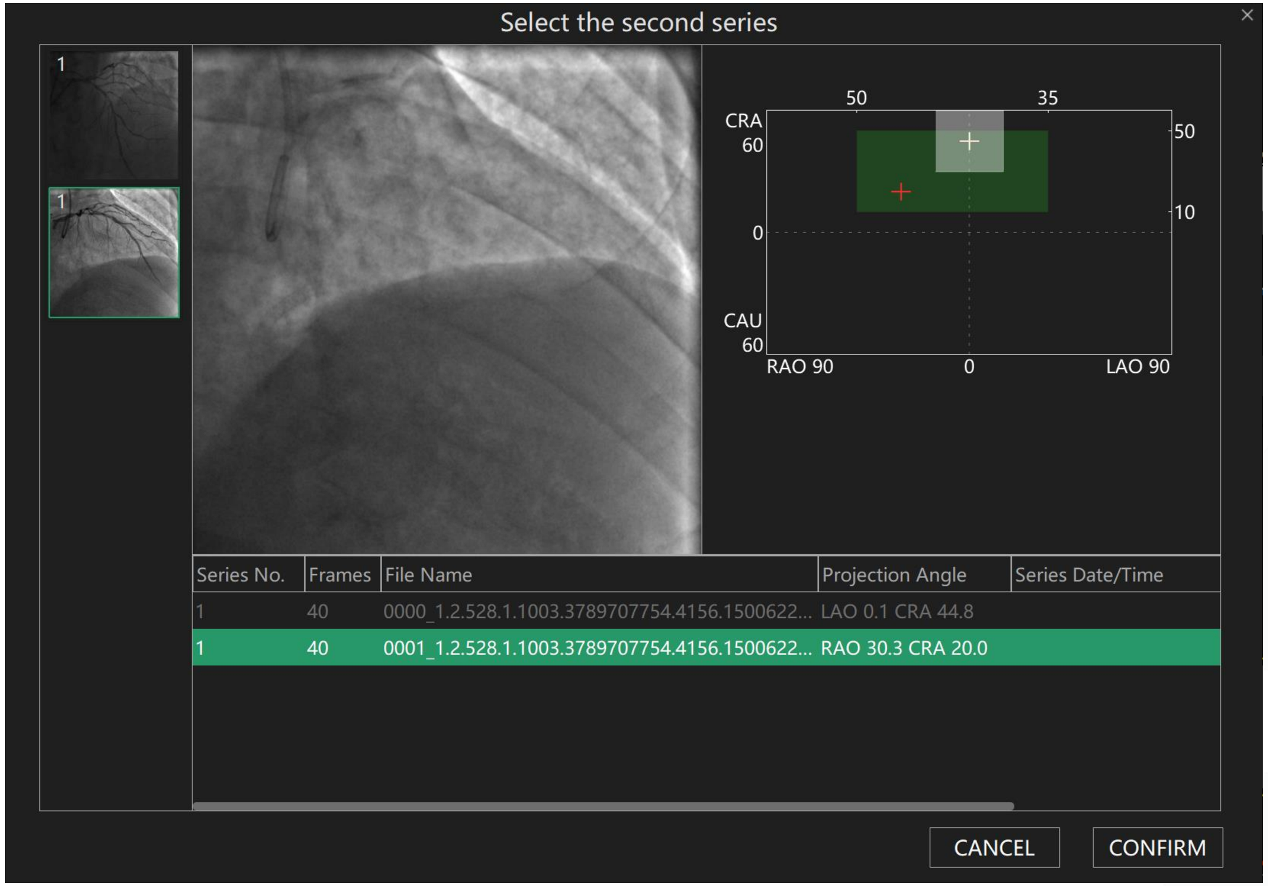
Note: After the 2D analysis, please click "DONE" and then click "Save” to **save the 2D results**. Otherwise, once starting 3D analysis, it will be impossible to save the 2D results again.

Second Image Selection	• **Recommended angle difference ≥25°,** minimal difference ≥15°, clear angiography, less overlapping, less foreshortening
Key Frame	• Prioritize the frame with the maximal stenosis exposure • Ensure the key frames of two selected image are in the same phase of the cardiac cycle
Coregistration	• **Start point and end point correspond to 1st image** • Check the alignment of the two images • Corresponding Marks
Check 3D Result	• Check 3D reconstruction morphology, reference lumen, pressure pull back curve

3D Analysis - Second Image Selection

Image Quilty: optimal angiography (especially in lesion segment) with less overlapping, less foreshortening.

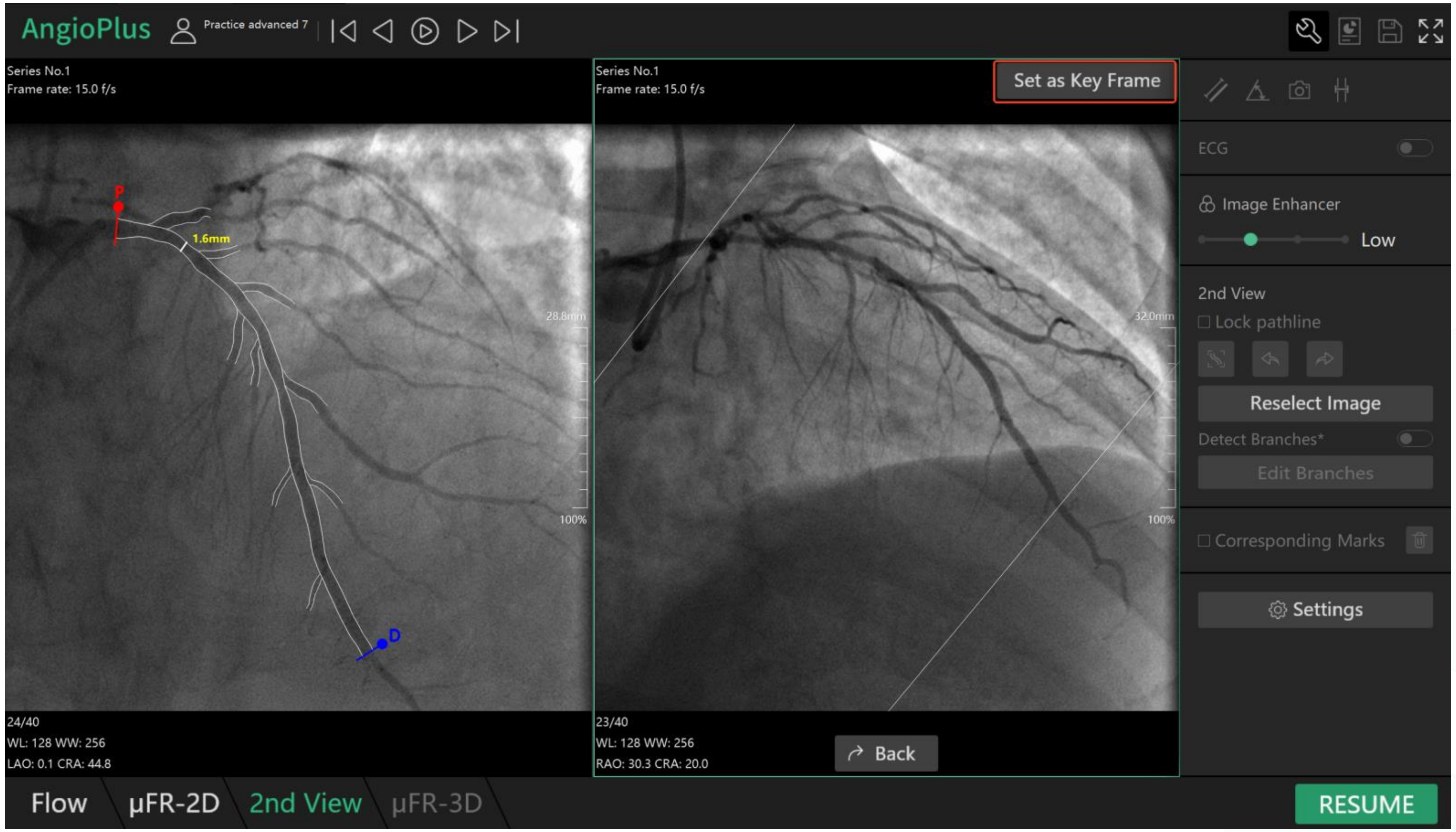
Second Image Angle: recommended angle difference ≥25°, minimal difference ≥15°.



3D Analysis - Key Frame

Ensure the key frames of two selected image are in the same phase of the cardiac cycle.

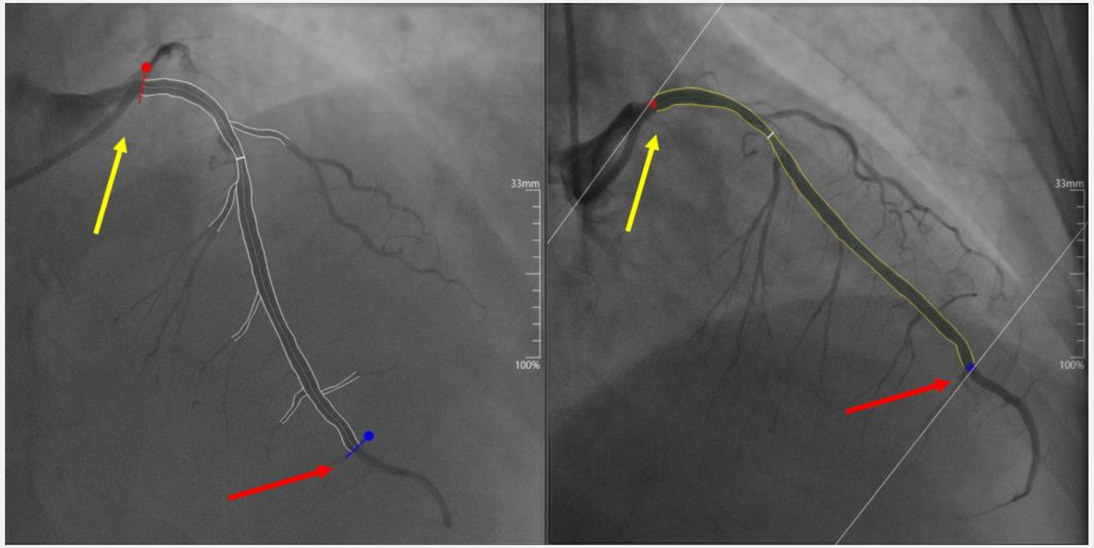
A. Maximal stenosis exposure > B. Lower foreshortening = C. Entire vessel clearly visualized > D. Less overlapping

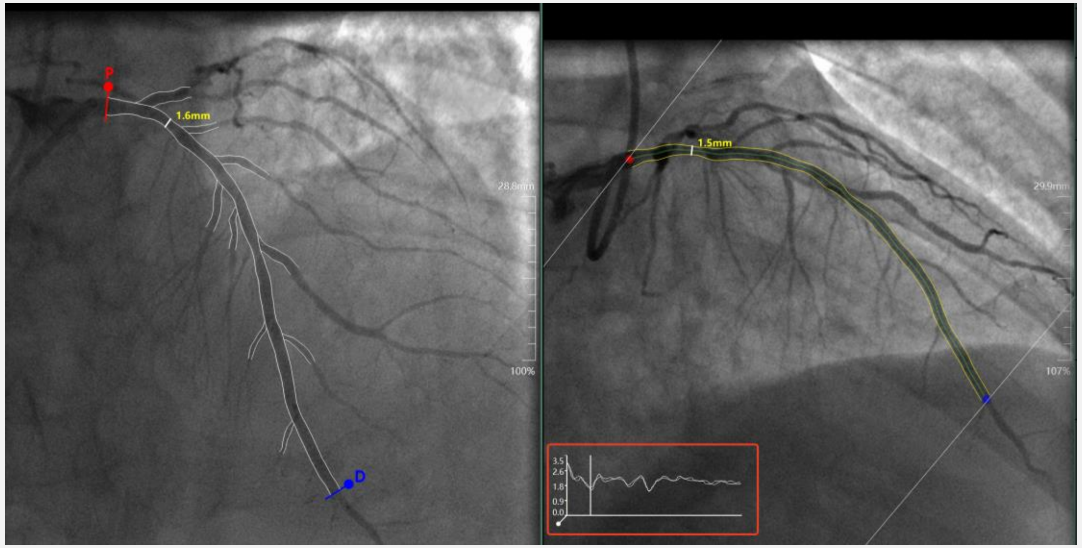


3D Analysis - Coregistration

ROI: choose the same start point and end point with the 1st image.

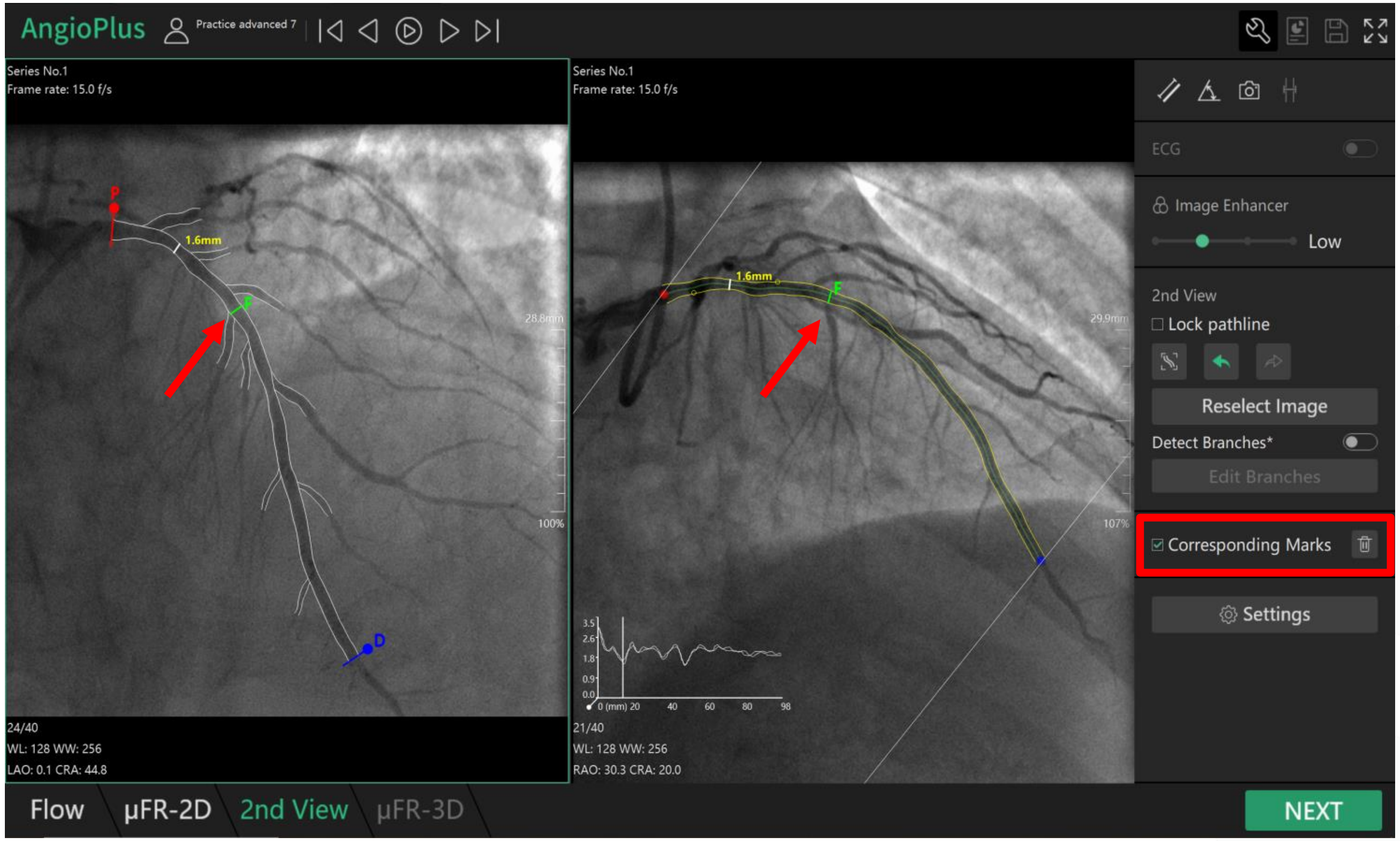
Alignment: double check the alignment (especially at the most severe stenosis) and corresponding curve. If the contour is incorrectly identified, moderately adjust the contour of the second image.





3D Analysis - Coregistration

Corresponding Marks: If the alignment starts and ends correctly, but **the middle part** still cannot be matched, especially for the segment around the lesion, use the “corresponding marks” function to prioritize the alignment of the most severe stenosis.



3D Analysis - Check 3D Result

Check 3D reconstruction morphology, reference lumen, pressure pull back curve.

Double click to change displayed angiography image

Demonstration

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